

Segmented inversion of the Gaussian correction and large-coefficient precision

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Setting

This uses PR #5's chirped-correlation lemma for the correlations and its parameter conventions. PR #5 is unmerged on `CrocSwap/integer-mult-bounds`.

Keep the notation of the one-dimensional resampling lemma: $2 \leq s < t$, $\sigma = t/s$, $\theta = \sigma - 1 \leq 1/4$, $\alpha \geq 2$ with $\alpha^2\theta \geq 1$, $q_j = \lfloor tj/s + 1/2 \rfloor$, $\beta_j = tj/s - q_j \in [-1/2, 1/2)$, defined for all integers j , so $q_{j+s} = q_j + t$ and $\beta_{j+s} = \beta_j$. The matrix $N = CTD$ acts on s -periodic sequences by

$$(Nu)_\ell = \sum_{j \in \mathbb{Z}} e^{-\pi\alpha^2 X_{\ell,j}} u_j, \quad X_{\ell,j} = (\sigma j - q_\ell)^2 - \beta_j^2.$$

1 Exact structure

Lemma 1 (Closed form). *Put $n = q_j - q_\ell \in \mathbb{Z}$. Then $X_{\ell,j} = n(n + 2\beta_j)$.*

Proof. $\sigma j - q_\ell = (q_j + \beta_j) - q_\ell = n + \beta_j$, so $X = (n + \beta_j)^2 - \beta_j^2 = n(n + 2\beta_j)$. $\square$

Since $\sigma = 1 + \theta$ with $\theta < 1$, consecutive values satisfy $q_{j+1} - q_j \in \{1, 2\}$. Call j a *wrap* when the difference is 2; between wraps $\beta_{j+1} = \beta_j + \theta$. A *segment* is a maximal run of indices containing no wrap step. Each segment has $\lfloor 1/\theta \rfloor$ or $\lceil 1/\theta \rceil$ indices, up to one boundary index, and wraps have density $t/s - 1 = \theta$.

Lemma 2 (Chirp-Toeplitz segments). *Let ℓ, j lie in one segment and choose the real centre $\ell^* = \ell - (\beta_\ell + 1/2)/\theta$, which is the same for the whole segment. With $u = \ell - \ell^*$, $v = j - \ell^*$ and $h = j - \ell$,*

$$e^{-\pi\alpha^2 X_{\ell,j}} = e^{\pi\alpha^2\theta u^2} \tau_h e^{-\pi\alpha^2\theta v^2}, \quad \tau_h = e^{-\pi\alpha^2(\sigma h^2 - h)}.$$

Thus the principal block M_k of N on segment k equals $D_k T_{L_k} D_k^{-1}$, a similarity, where T_L is the $L \times L$ Toeplitz matrix with symbol (τ_h) , which does not depend on the segment, and $D_k = \text{diag}(e^{\pi\alpha^2\theta u^2})$.

Proof. Inside a segment $n = h$ and $\beta_j = \beta_\ell + \theta h = \theta v - 1/2$. By Lemma 1, $X = h(h + 2\theta v - 1)$. Using $2hv = v^2 - u^2 + h^2$ gives $X = (1 + \theta)h^2 - h + \theta v^2 - \theta u^2$, which is the stated factorisation. $\square$

Lemma 3 (Cross-wrap entries). *If ℓ, j lie in different segments then $|n| \geq |h| + 1 \geq 2$ and $X_{\ell,j} \geq |n|(|n| - 1) \geq 2$. In particular every cross-segment entry is at most $e^{-2\pi\alpha^2}$. An entry exceeds 2^{-Q} only if $|n|(|n| - 1) < Q/(\pi\alpha^2 \log_2 e)$.*

Proof. Crossing $k \geq 1$ wrap steps adds k to $|q_j - q_\ell|$ beyond $|h|$. Lemma 1 with $|\beta_j| \leq 1/2$ gives $X \geq |n|(|n| - 1)$. $\square$

The identities in Lemmas 1–3 were checked in exact rational arithmetic for five (s, t) pairs from $(113, 128)$ to $(4093, 4096)$, over three periods and $|j - \ell| \leq 30$: no mismatches, and the minimum cross-segment exponent was $228/113$ and $2020/1009$.

Indexing. Indices live in $\mathbb{Z}/s\mathbb{Z}$. Index 0 has $\beta_0 = 0$, so it lies mid-segment, and one segment always crosses the period boundary; on tape it is stitched from the two line ends. Aliased terms $j + ms$ belong to Δ , not to M_k . They matter only if one segment covers a whole period ($t - s = 1$), which the parameters exclude.

2 Applying N^{-1}

Write $N = M + \Delta$ with $M = \bigoplus_k M_k$. Fix a target accuracy 2^{-Q} and drop entries of Δ below $2^{-Q - \log_2 s - 4}$. By Lemma 3, what remains of Δ sits in $w \times w$ corner blocks at each wrap, with

$$w \leq 1 + \sqrt{Q/(4.53\alpha^2)}.$$

Proposition 1 (Segmented inverse). *Assume $\|N - I\| < 1/2$, which holds when $\alpha^2\theta \geq 1$. Then $N^{-1}u$ can be computed to absolute error 2^{-Q} , for $\|u\| \leq 1$, using*

1. two applications of M^{-1} ;

2. per wrap, $O(w^3)$ arithmetic operations at precision P to form and factor the capacitance block, and $O(w^2)$ to apply it;

with working precision $P = Q + \lceil 10.2 \alpha^2 / \theta \rceil + O(\log s)$ when whole segments are used as blocks. With the recentred sub-blocks of the last section, the reserve is at most Q instead. Each application of M^{-1} costs a constant number of chirped correlations of segment length, hence $O(P^{1+\delta})$ per output. The setup computes the Gohberg–Semencul vectors of T_L and T_{L+1} once per line, in $O(L^2)$ operations.

Proof sketch. Segment blocks. M_k is a principal submatrix of $N = I + E$, so $\|M_k - I\| \leq \|E\| < 1/2$ and $\|M_k^{-1}\| \leq 2$. Also $\|T_L - I\| \leq \sum_{h \neq 0} \tau_h < 2e^{-\pi\alpha^2\theta} < 0.1$, so T_L is well conditioned and has a Gohberg–Semencul representation $T_L^{-1} = x_0^{-1}(L(x)U(y) - L(z)U(w))$ with triangular Toeplitz factors. Applying $M_k^{-1} = D_k T_L^{-1} D_k^{-1}$ means: scale by $e^{-\pi\alpha^2\theta u^2} \leq 1$, perform four triangular Toeplitz products (correlations, evaluated as in the chirped-correlation lemma), then scale by $e^{\pi\alpha^2\theta v^2} \leq e^{\pi\alpha^2\theta(L+1)^2}$. Since $L \leq 1/\theta + 1$, the last factor is at most $2^{10.2\alpha^2/\theta}$ for $\theta \leq 1/4$ (and about $2^{4.53\alpha^2/\theta}$ as $\theta \rightarrow 0$), and this is the precision reserve in P . The exact result has norm at most $2\|u\|$.

Coupling. Write $\Delta = UV^T$, where V selects the $2w$ columns around each wrap and U holds the corresponding columns of Δ . Woodbury gives

$$N^{-1} = M^{-1} - M^{-1}U C^{-1}V^T M^{-1}, \quad C = I + V^T M^{-1}U.$$

C is cyclic block tridiagonal with $2w \times 2w$ blocks, because M^{-1} is block diagonal, each block of U touches the two segments adjacent to its wrap, and indices are taken modulo s . This needs $2w < L$, which holds for $d \geq 3$. Solve it by bordered (cyclic) block elimination, still $O(w^3)$ per wrap, with one forward and one backward sweep. Its entries need only the corner entries of each M_k^{-1} . From the Gohberg–Semencul form, entry (a, b) is a sum of $\min(a, b) + 1$ products. For the bottom-right corner use persymmetry, $(T^{-1})_{L-1-a, L-1-b} = (T^{-1})_{b, a}$, so every corner entry costs $O(w)$, and $O(w^3)$ per wrap. Also $\|C - I\| \leq \|M^{-1}\| \|\Delta\| \leq 4we^{-2\pi\alpha^2}$, so block elimination without pivoting is stable, at $O(w^3)$ per wrap. $V^T M^{-1}u$ reads entries of the first $M^{-1}u$. The vector Uz is supported near wraps, and the second M^{-1} application handles it. $\square$

Remark (Cost per output). Wraps have density θ . Per output the coupling therefore costs $O(\theta w^3)$ operations at precision P .

3 Large coefficients

Split the inputs into b' -bit digits instead of $b = \lceil \log_2 n \rceil$ -bit digits, with $b' = \lceil b^{1+x} \rceil$ for a fixed $x > 0$, and set $p = 6b'$. Then $T \in [4n/b', 8n/b']$, the volume is $Tp = \Theta(n)$, and the address length is $\log_2 T = b - (1+x)\log_2 b + O(1)$. Choose $d = \lfloor b^\epsilon \rfloor$ and r, ℓ from T as before. The capacity bound becomes $c_j < 2^{3b'}$, and the final error bound becomes

$$2^{2b'+4} S \cdot 28 \cdot 2^\gamma T^2 L \cdot 2^{-6b'} < 448 b' 2^{-b'+\gamma} < \frac{1}{2} \quad \text{whenever } \gamma \leq b'/4.$$

With $\alpha^2 = \lfloor b'/(8d) \rfloor$ we have $\gamma = 2d\alpha^2 \leq b'/4$, and with $\theta \approx 1/(4dL^y)$, $\alpha^2\theta \geq b'/(32d^2L^y) \geq 1$ once $2\epsilon + y < 1 + x$. All uses of $r \geq 2^{a_r p/d}$ in the layer and transform proofs only need r to outgrow every fixed power of p . Here $r = 2^{\Theta(b^{1-\epsilon})}$ and $p = \Theta(b^{1+x})$, so that still holds.

Cutoffs. Every “for large L ” in this note and in the stack notes is asymptotic only: with $\epsilon = 0.9999$ the axis width is $\ell = L^{10^{-4}}$, and conditions such as $\ell - 1 > 3G$ or $\lambda^2 \leq s/p$ hold only for astronomically large L . This is enough for the $O(\cdot)$ statement and gives no concrete range.

4 Gaussian row

With whole segments as blocks, $w^2 = O(d)$ and the oversampling $\theta \approx 1/(4d)$ give $\theta w^3 = O(d^{1/2})$ coupling work per output. With the recentred sub-blocks and reduced oversampling of the next section, the coupling work is $O(1)$ per output. The Gaussian row is then $O(dp^\delta)$, with exponent $\epsilon + (1+x)\delta$ measured in powers of $\log n$. This is the row used in the certificate.

5 Sub-blocks with recentred chirps

Lemma 4 (Recentred blocks). *Let ℓ, j and a centre c lie in one segment. Put $u = \ell - c$, $v = j - c$ and $h = j - \ell$. Then*

$$X_{\ell,j} = \sigma h^2 + 2\beta_c h + \theta(v^2 - u^2).$$

Hence the principal block of N on any run of consecutive indices inside a segment starting at c equals $D_c T_c D_c^{-1}$, with $D_c = \text{diag}(e^{\pi\alpha^2\theta u^2})$ and T_c the Toeplitz matrix with symbol $e^{-\pi\alpha^2(\sigma h^2 + 2\beta_c h)}$.

Proof. In a segment $\beta_\ell = \beta_c + \theta u$ and $\beta_j = \beta_c + \theta v$. Expand $X = (\sigma h + \beta_\ell)^2 - \beta_j^2$, use $\sigma - \theta = 1$ and $2hu = v^2 - u^2 - h^2$. $\square$

This was checked in exact rational arithmetic for three centres per pair on all five test cases.

One Toeplitz inverse serves all blocks. For $b \in [-1/2, 1/2]$ let $T^{(b)}$ be the $\lambda \times \lambda$ Toeplitz matrix with symbol $e^{-\pi\alpha^2(\sigma h^2 + 2bh)}$, so $T_c = T^{(\beta_c)}$ and $T_\lambda = T^{(-1/2)}$. With $\rho_c = e^{-\pi\alpha^2(2\beta_c + 1)} \in (0, 1]$ and $G_c = \text{diag}(\rho_c^{-i})$ we have $T_c = G_c T_\lambda G_c^{-1}$, so $(T_c^{-1})_{ab} = \rho_c^{b-a} (T_\lambda^{-1})_{ab}$.

Lemma 5 (Envelope). *Let $|b| \leq 1/2$, $\alpha^2 \geq 4$, $\alpha^2\theta \geq 1$, $\mu_\pm = e^{-\pi\alpha^2(\sigma \pm 2b)} \leq e^{-\pi\alpha^2\theta}$ and $c_\pm = 2\mu_\pm \leq 2e^{-\pi}$. Then $|(T^{(b)})_{ij}^{-1}| \leq \mathcal{E}(j-i)$, where $\mathcal{E}(0) = 1.12$, $\mathcal{E}(h) = 2.0001 c_+^h$ for $h > 0$ and $\mathcal{E}(h) = 2.0001 c_-^{|h|}$ for $h < 0$.*

Proof. $\|T^{(b)} - I\|_\infty \leq \mu_+ + \mu_- + \sum_{|h| \geq 2} e^{-\pi\alpha^2(\sigma h^2 - |h|)} < 0.087$, which gives the bound $1/(1 - 0.087) < 1.12$ for every entry. For $h = j - i > 0$ conjugate by $R = \text{diag}(c_+^i)$: $RT^{(b)}R^{-1}$ has entries $t_h c_+^{-h}$, whose off-diagonal row sums are at most $\frac{1}{2} + \sum_{h \geq 2} 2^{-h} e^{-\pi\alpha^2\sigma h(h-1)} + \sum_{m \geq 1} 2^m e^{-\pi\alpha^2\sigma m(m+1)} < \frac{1}{2} + 10^{-10}$. So $\|(RT^{(b)}R^{-1})^{-1}\|_\infty < 2.0001$ and $|(T^{(b)})_{ij}^{-1}| \leq 2.0001 c_+^h$. The case $h < 0$ is the same with $R = \text{diag}(c_-^i)$. $\square$

Gohberg–Semencul vectors. Put $x_c = T_c^{-1}e_0$, $y_c = T_c^{-1}e_{\lambda-1}$, J the reversal and Z the down shift. Then $T_c^{-1} = x_{c,0}^{-1}(L(x_c)U(Jy_c) - L(Zy_c)U(ZJx_c))$, with $x_{c,0} \in [0.9, 1.1]$. Every entry of the four vectors is an entry of T_c^{-1} , so Lemma 5 bounds it, and each triangular factor has ∞ -norm at most $\sum_h \mathcal{E}(h) \leq 1.31$. The partial products need no cancellation: for all a, b ,

$$\sum_k |x_{c,a-k}(Jy_c)_{b-k}| \leq 1.13 \mathcal{E}(b-a), \quad \sum_k |(Zy_c)_{a-k}(ZJx_c)_{b-k}| \leq 4c_+c_- \mathcal{E}(b-a) \leq 16e^{-2\pi\alpha^2} \mathcal{E}(b-a),$$

because the second sum is dominated by $|(T_c^{-1})_{a-1,\lambda-1}(T_c^{-1})_{\lambda-b,0}|$. So $L(Zy_c)U(ZJx_c)$ is a correction of relative size $e^{-2\pi\alpha^2}$ to $L(x_c)U(Jy_c)$, and both obey the envelope of T_c^{-1} itself.

Rescaling. The vectors of T_c are those of T_λ rescaled by G_c as whole columns: $x_{c,i} = \rho_c^{-i}x_i$ and $y_{c,i} = \rho_c^{\lambda-1-i}y_i$. Conjugating each triangular factor by G_c separately gives the same product, but moves a factor $\rho_c^{\mp\lambda}$ into $L(Zy)$ and $U(ZJx)$. That factor is up to $2^{9.06\alpha^2\lambda}$, which a fixed-point correlation would have to carry as extra bits. Under the parameters of the last section $9.06\alpha^2\lambda \approx 1.23 QL^{y/2}$, which is not $O(p)$, so the column rescaling is required.

Storage in two frames. Store $x^+ = (T^{(1/2)})^{-1}e_0$ and $y = T_\lambda^{-1}e_{\lambda-1}$, both bounded by Lemma 5, in fixed point with absolute precision 2^{-P} . Then

$$x_{c,i} = r_c^i x_i^+, \quad r_c = e^{-\pi\alpha^2(1-2\beta_c)} \leq 1, \quad y_{c,i} = \rho_c^{\lambda-1-i} y_i, \quad \rho_c \leq 1.$$

Every rescaling factor is at most 1, so it only contracts stored errors. Storing T_λ 's own x in fixed point instead loses $\min(2^{-P}, |x_i|)$, which ρ_c^{-i} amplifies by up to $2^{9.06\alpha^2 i}$. Storing it in floating point avoids that loss, but the setup then needs absolute precision $P + 9.06\alpha^2\lambda$.

Error. Form r_c^i and ρ_c^i by repeated multiplication in fixed point, so every vector entry has error at most $(\lambda + 1)2^{-P}$. Let the four correlations round to 2^{-P} . Then for $\|z\|_\infty \leq 1$, the computed $T_c^{-1}z$ is within $7(\lambda + 1)^2 2^{-P}$. Applying $M_c^{-1} = D_c T_c^{-1} D_c^{-1}$ scales the input by $D_c^{-1} \leq 1$ and the output by at most 2^{R_D} , with $R_D = 4.53\alpha^2\theta(\lambda - 1)^2 \leq Q$. The block error is therefore at most $7(\lambda + 1)^2 2^{R_D - P} \|v\|_\infty$. Corner entries for the capacitance are single entries of the same form, with error at most $7(\lambda + 1)2^{-P}$. Hence

$$P = 2Q + 2\log_2(\lambda + 1) + \log_2 s + 8 = O(p)$$

suffices, with $\lambda = O(dL^{y/2})$ under the parameters of the last section.

Cuts. Cut every segment into near-equal blocks of length between $2w$ and λ , and treat each cut like a wrap.

- Coupling entries across a cut have $n = h$ and $X = h(h + 2\beta_j) \geq |h|(|h| - 1)$, which is at least 2θ for $h = \pm 1$. So they are at most $e^{-2\pi\alpha^2\theta} \leq e^{-2\pi}$, and only entries with $|h| \leq w$ matter.
- All block solves use principal blocks of $N = I + E$, hence $\|M^{-1}\| \leq 2$.
- With $\|\Delta\| \leq e^{-2\pi\alpha^2\theta} + 2.01e^{-2\pi\alpha^2}$, we get $\|C - I\| \leq \|M^{-1}\|\|\Delta\| < 0.005$. Also $C^{-1} = I - V^\top N^{-1}U$ gives $\|C^{-1}\| \leq 1 + \|N^{-1}\|\|\Delta\| < 1.01$.
- The capacitance is cyclic block tridiagonal, with one block per cut or wrap, because every block has length at least $2w$. Eliminate without pivoting.

Precision and cost. Inside a block, the chirp D_c spans at most $e^{\pi\alpha^2\theta\lambda^2}$, that is $4.53\alpha^2\theta\lambda^2$ bits. Choose

$$\lambda = \left\lfloor \sqrt{Q/(4.53\alpha^2\theta)} \right\rfloor,$$

so that the D_c precision reserve is at most Q . The working precision is $P = 2Q + O(\log s) = O(p)$. Couplings occur with density $1/\lambda + \theta$ per output, each costing $O(w^3)$ with $w^2 = O(Q/\alpha^2)$.

Replaced hypothesis. Upstream's resampling lemma assumes $\theta > p/\alpha^4$, which is used only to get $\alpha^2\theta > 1$ and to bound its Neumann count. Here the Neumann series is gone, so that hypothesis is replaced by $\alpha^2\theta \geq 1$. PR #5's lemmas, which assume the hypotheses of the upstream lemma, are used under this replacement. Every remaining use of $\alpha^2\theta \geq 1$ is $\|N - I\| < 2.01e^{-\pi\alpha^2\theta/2} < 0.42$.

Reduced oversampling. Take $\theta_i \approx \eta = 1/(4dL^y)$ in prime selection (Section 1 of the stack notes), so the padding factor is $\prod_i(1 + \theta_i) = 1 + o(1)$. Take $\alpha^2 = \lfloor Q/(48d) \rfloor$ with $Q = \Theta(p)$. Then $\gamma = 2d\alpha^2 \leq b'/4$, and $\alpha^2\theta \geq 1$ once $Q \geq 192d^2L^y$, which holds when $2\epsilon + y < 1 + x$. This gives $w^2 = O(d)$ and $\lambda = \Theta(dL^{y/2})$, so the coupling costs $O(w^3(1/\lambda + \theta)) = O(d^{1/2}L^{-y/2} + d^{1/2}L^{-y}) = O(1)$ per output for $y = \epsilon$. Including the $O(1)$ chirped correlations, each axis costs $O(p^{1+\delta})$ per output. The Gaussian row is therefore $O(dp^\delta)$: exponent $\epsilon + (1 + x)\delta$.

Tape schedule and setup. Process one line in increasing index order.

- For each block, compute β_c from the counters that already generate q_j , in $O(p)$ time.
- Form r_c^i and ρ_c^i for $0 \leq i \leq \lambda$ by repeated multiplication in fixed point at precision P . Both are at most 1. That is 2λ products, so $O(P^{1+\delta})$ per output.
- Rescale the two stored vectors x^+ and y into the four Gohberg–Semencul vectors of T_c , and apply the four triangular Toeplitz products as correlations of length λ in fixed point. Each block uses the packed multiplication of the chirped-correlation lemma, at $O(\lambda P^{1+\delta})$ per block.
- Write the block's corner entries ($O(w^2)$ numbers) to a capacitance tape in index order.
- The cyclic block-tridiagonal solve makes one forward sweep and one backward sweep over that tape, reading fixed-width records in each direction. The single cyclic border is carried on a separate work tape.
- A second pass applies M^{-1} to Uz , which is supported near cuts and wraps, and then subtracts.

All passes move the line tape $O(1)$ times, so the line costs $O(sP)$ in movement. Arithmetic costs $O(s(1 + w^3(1/\lambda + \theta))P^{1+\delta})$. Setup solves $T^{(1/2)}x^+ =$

e_0 and $T_\lambda y = e_{\lambda-1}$, and the same for λ' (the final short block of each segment), once per line at precision $P + O(\log \lambda)$. The Neumann iteration $x \leftarrow e + (I - T)x$ contracts by 0.087 per step, so $O(P)$ steps of $O(\lambda^2)$ operations suffice, in $O(\lambda^2 P^{2+\delta})$. This is negligible when $\lambda^2 P \leq s$, which holds for large L . No step needs more than P bits.

Numerical check. `scripts/check_subblock_inverse.py` cuts the segments of the (60, 67), $\alpha^2 = 9$ instance into blocks of length 2 and 3, solves each block by its own recentred Toeplitz system at 120 digits, couples all cuts and wraps by Woodbury, and matches the direct solve to $3 \cdot 10^{-64}$, against a target of 2^{-200} . `scripts/check_reuse_inverse.py` checks a dense stored inverse of T_λ rescaled per block, with a fixed-point negative control. `scripts/check_gs_path.py` tests this path itself: two stored vectors per block length in fixed point, contracting rescales, four fixed-point triangular products per block, and Woodbury coupling. On (241, 256) with $\alpha^2 = 17, 24$ and (113, 128) with $\alpha^2 = 8$, at $P = Q + R_D + 2 \log_2 \lambda + 8$ bits, it matches the direct solve to $2.1 \cdot 10^{-49}$, $3.6 \cdot 10^{-50}$ and $2.8 \cdot 10^{-35}$, against targets $7.0 \cdot 10^{-46}$ and $7.9 \cdot 10^{-31}$. It fails as predicted in two negative controls. Dropping the reserve R_D from P gives errors of 10^{-26} to 1. Storing T_λ 's vectors in fixed point gives $4.7 \cdot 10^{-10}$ and $1.1 \cdot 10^{-13}$ on (241, 256).